

## Manufacturing process:

Metal  
add<sup>n</sup>

Metal  
Removal

Bulk  
deformation

## Sand casting:

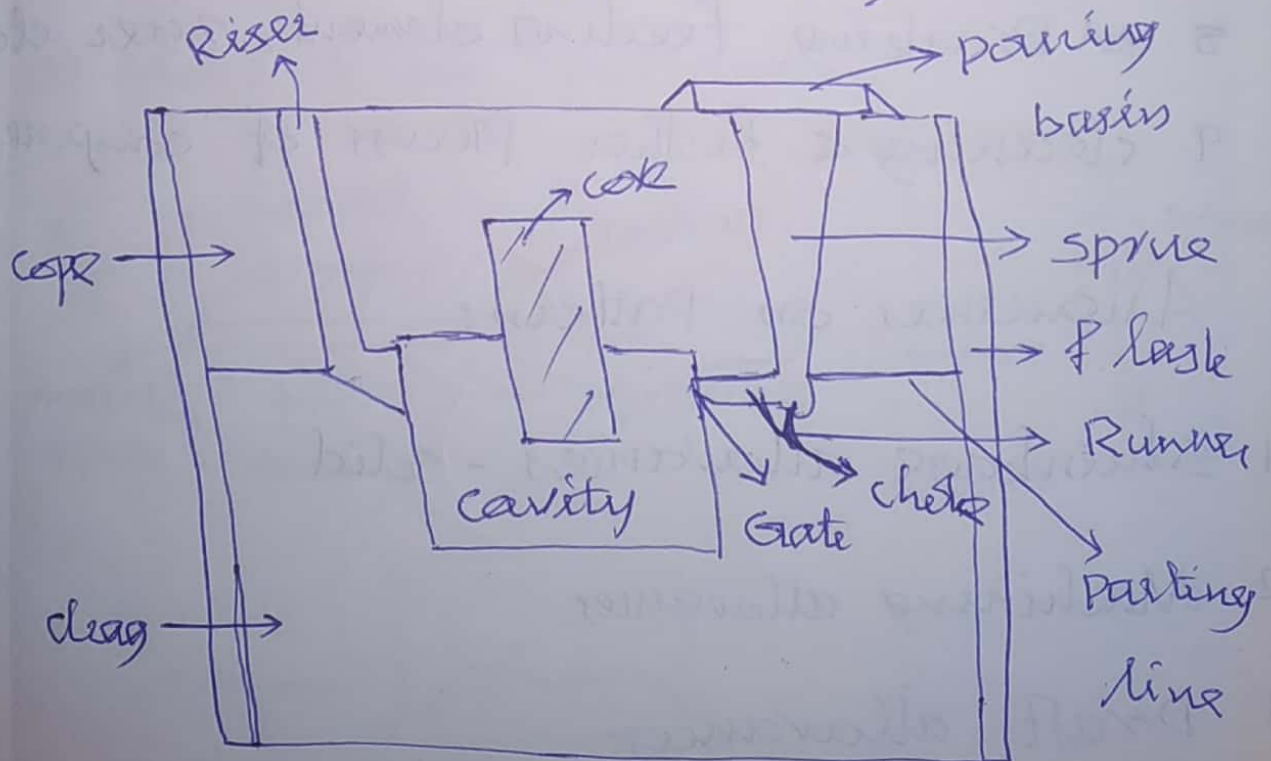
Cavity - It gives external feature

Coke - It gives internal feature

Coke box - to prepare coke

Riser -

Shrinkage  $\begin{cases} \text{liquid shrinkage} \\ \text{solid shrinkage} \end{cases}$



Conical shape of sprue is to reduce aspiration effect.

Steps:

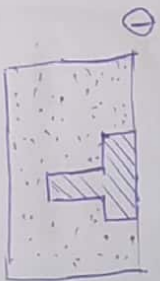
1. Pattern making
2. Preparing sand cavity by using pattern
3. Preparing core box with wood/metal
4. Preparing core by using core box
5. If needed, heating core
6. Melting & pouring molten metal
7. Cooling & breaking mold to free the component
8. ~~1~~ Breaking feeding element, riser etc
9. cleaning & further process of component

Allowances on Pattern:

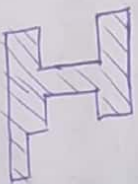
1. Shrinkage allowances - solid
2. Machining allowances
3. Draft allowances
4. Shape allowance

Types of Pattern:

1. Single pattern
2. Split pattern

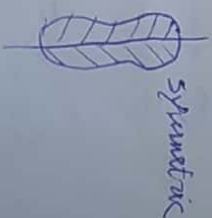
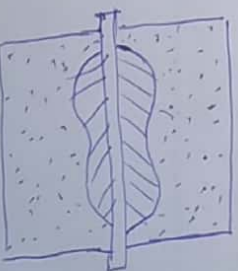
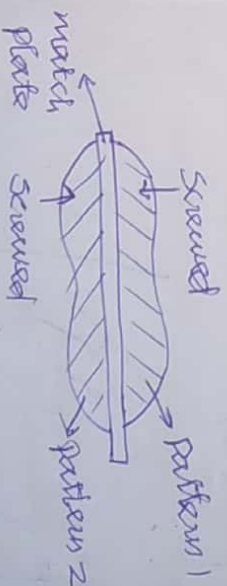


③ ~~Loss~~ Piece pattern:

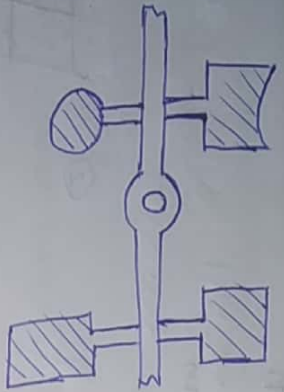


→ separately made

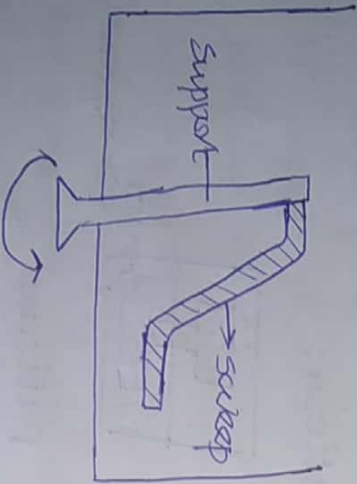
④ Match Plate pattern:



### ⑤ Gates Pattern:

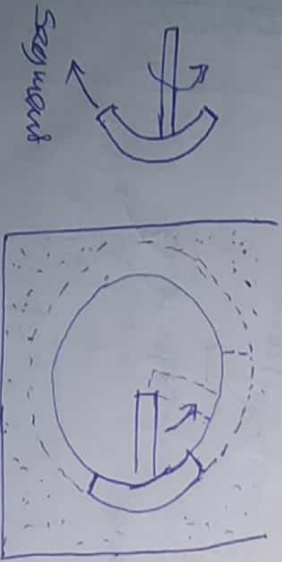


### ⑥ Sweep pattern:

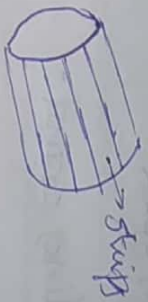


rotating the support  
& get a complete  
shape.

### ⑦ Segmental Pattern:



### ⑧ Skeleton pattern:



### colour coding for patterns:

Red - This portion should be left as it is

Black - portion of coke

Yellow - surface need machining

Green - loose piece pattern portion

### Elements of sand casting:

→ Refractory sand grains:

eg: silica, silicon, coke, carbon, Dolomite

→ Binders:

eg: clay, organic / phenolic resins (epoxy)

→ Facing material:

eg: sea coal

silica flour



→ Cushion:

eg: wood, flour, coal, etc.

Properties of Molding sand:

- 1 Refractoriness
  - 2 Permeability
  - 3 collapsibility
  - 4 Green sand strength
  - 5 Flowability
  - 6 cohesiveness & Adhesiveness
- \* Natural sand - 21% clay, 65% silica  
rest impurities
- \* Synthetic sand - clay can be limited to 5%  
uniform size, more refractoriness.

Testing of sand:

1 Moisture Test:

2 Green strength (in rammed condition)

3 Density test

4 Dry compression strength

Types of sand Moulds:

1 Green sand Mould:

- moderate strength suitable for cavity
- good collapsibility
- good permeability
- least expensive
- moisture can cause defects

2 Dry sand Mould:

- better dimensional accuracy
- suitable for core
- expensive
- Time consuming
- poor collapsibility

3 Skirted Mould:

- have advantage of both green, dry mould
- moderately expensive due to higher labour charge than green
- No core locating frames needed

# Cores:

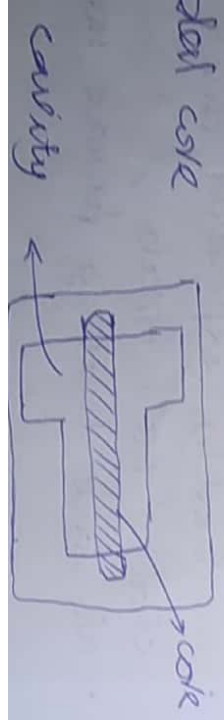
- made by sand, binders, mixers in a core box
- synthetic core binders like urea phenolmelaldehyde are common for low temp metals.
- Phenol phenolmelaldehyde for high temp metals.

## core characteristics:

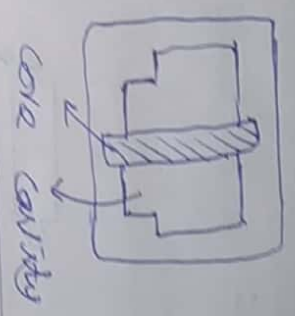
- \* good dry strength after baking
- \* good green strength to retain the shape before baking
- \* refractoriness, surface smoothness
- \* low gas formation, permeability

## Types of core - Based on design:

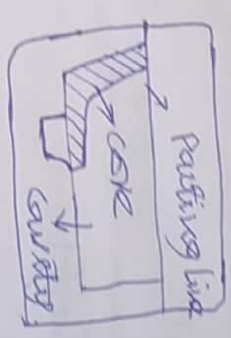
### Horizontal core



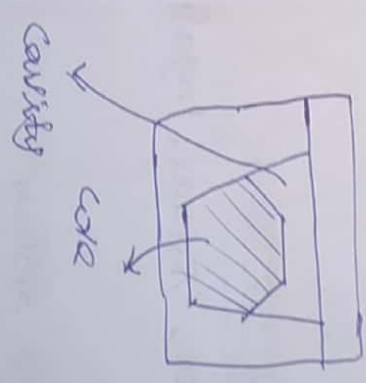
### 2 Vertical core:



### 4 Drop/stop off core:



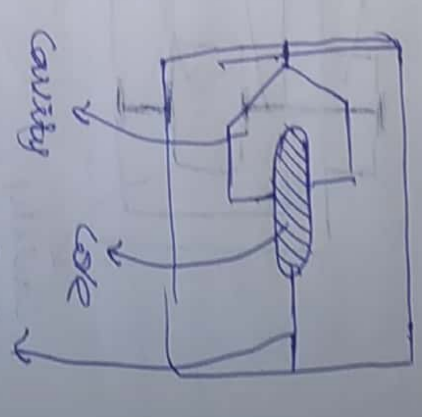
### 6 Rain up core:



### 3) Hanging core



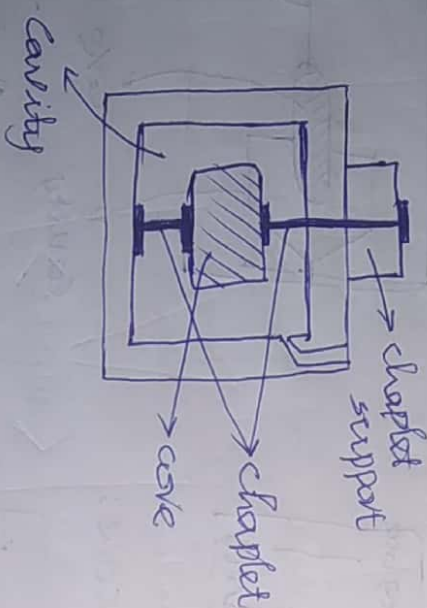
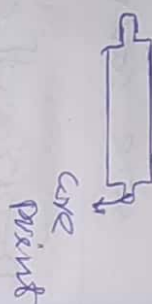
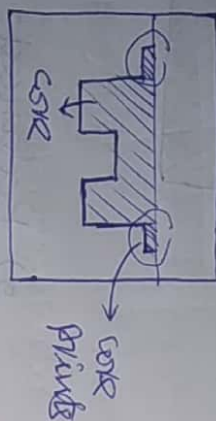
### 5) Balanced:





## Core supporting methods:

core prints & chaplets:



## Elements of Gating Systems:

Pouring basin, sprues, runners, ingate,

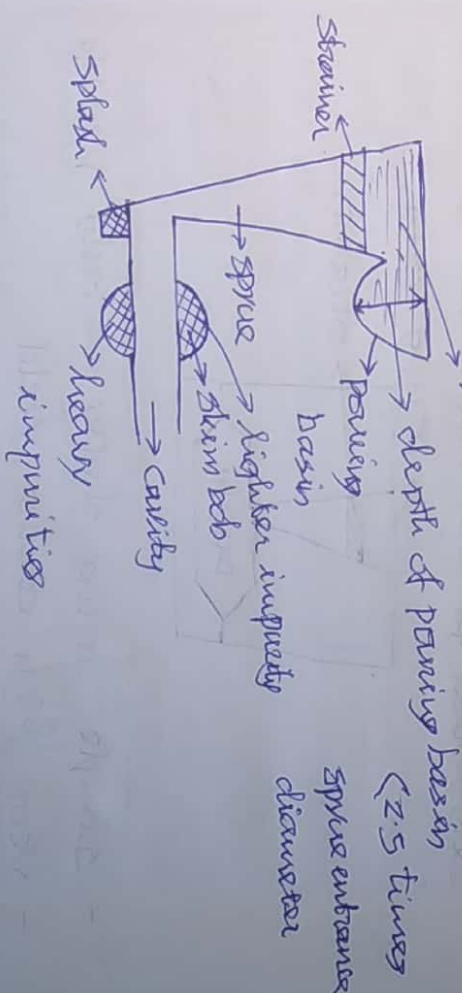
riser.

Characteristics of <sup>ideal</sup> gating system

\* mold should be filled in shortest possible

- \* time
- \* Turbulence should be minimum
- \* unwanted material like slag, dross should be eliminated
- \* Application of atmospheric air should be avoided
- \* Preheat thermal gradient
- \* No mold erosion
- \* economical & high casting yield

## Pouring Basin:



strainer core / delay core:

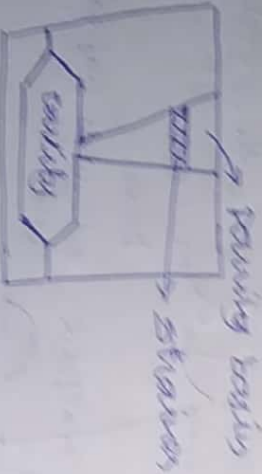
- \* small perforated screen, delays metal also remove dross

spoke core - reduce cooling time  
skin bf - traps impurities

## Types of gates:

### Top gate:

- since hotter metal is at top, favourable temp gradient possible - fast filling & weld
- since metal falling from height, weld erosion happens
- supported by run frames above



- simple casting design are only possible
- very high casting yield

### Bottom gate:

- very low mold erosion
- used for very deep molds

- very unfavourable temp gradient
- difficult preparation
- large sized iron needed

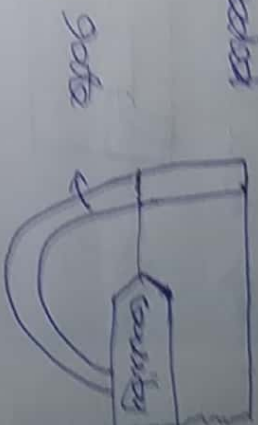
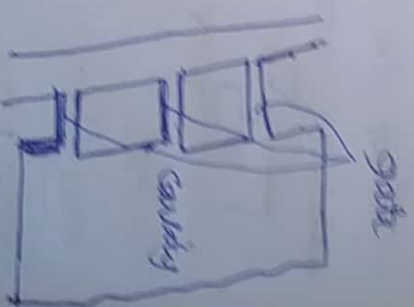
### Parting gate:

- wetten, metal entry is generally with parton
- moderate weld erosion
- moderate temp operations
- easy to make
- mold erosion is likely to be only on one side (bottom, cavity)



### Step gate:

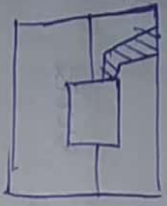
- heavy & large casting
- gradual filling without sand erosion





## Riser:

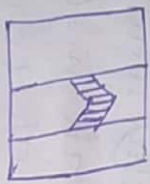
1. Top riser / open riser:



2. Blind riser:



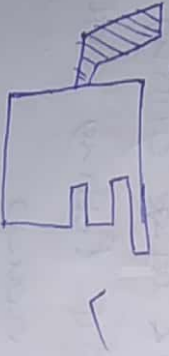
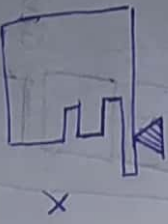
3. Internal riser:



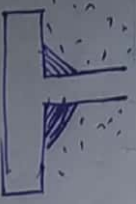
Directional Solidification:

1. Riser:

Place in such a way that volume to surface area is higher



2. Chills:



- external riser



- Internal riser

Riser size & solidification time:

Solidification time  $t = C \left( \frac{V}{S.A} \right)^2$

$V$  - Volume of casting

$S.A$  - surface area

$C$  - const which depends on type of metal

Pouring temp, mold material

Modulus method:

Stokes (modulus)  $\text{ratio} = 1.15 - 1.2$  (mod casting)

ie  $\left( \frac{V}{S.A} \right)_{\text{riser}} = 1.2 \left( \frac{V}{S.A} \right)_{\text{casting}}$

Caine's Relation:

Freezing ratio  $X = \frac{(S.A/V)_{\text{casting}}}{(S.A/V)_{\text{riser}}}$

Caine's  $\rightarrow X = \frac{a}{Y-b} + c$

$Y = \frac{(Vol)_{\text{riser}}}{(Vol)_{\text{casting}}}$

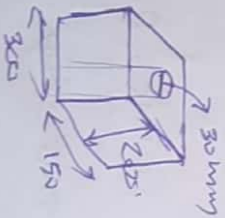


## Assignment

- freezing cast
- liquid-solid contraction cast
- relative freezing rate of riser say

? What are different equipments & machines which are used for preparing a ~~stone~~ sand cast? Module-2

? The dimension of a Al casting height is 200 mm, width 300 mm, depth 150 mm with a hole in centre which is having the diameter 30 mm. what will be dimension of wooden pattern if the linear allowance for Al is 0.013 mm/mm.



$$\begin{aligned} \text{300 mm} & \\ 300 + (300 \times 0.013) & \\ \text{200 mm} & \\ 200 + (200 \times 0.013) & \\ \text{150 mm} & \\ 150 + (150 \times 0.013) & \\ d = 30 \text{ mm} & \\ 30 - (30 \times 0.013) & \end{aligned}$$

? Dimension of Mg alloy casting is same as above what will be dimension of wooden pattern if pattern which is used for casting is to be made of Al. Assume double allowance for Mg 0.016 mm/mm for Al 0.013 mm/mm

- Al pattern dimension:

$$\begin{aligned} 200 + (200 \times 0.016) &= 203.2 & 30 - (30 \times 0.016) \\ 150 + (150 \times 0.016) &= 152.4 & = 29.52 \\ 300 + (300 \times 0.016) &= 304.8 & \\ \text{wooden pattern dimension:} & & \\ 203.2 + (203.2 \times 0.013) &= 205.8416 & \\ 29.52 + (29.52 \times 0.013) &= 29.13 & \\ 152.4 + (152.4 \times 0.013) &= 154.38 & \\ 304.8 + (304.8 \times 0.013) &= 308.76 & \end{aligned}$$

? A spherical ~~cast~~ <sup>ball</sup> made of metal of metal having radius 2 mm solidifies in 10 sec. what will be the solidification time for a similar metal ball having radius 4 mm

as per the chakraverty rule.

$$t = k \left( \frac{V}{5 \cdot A} \right)^2$$

$$\frac{t_1}{t_2} = \frac{\left( \frac{V_1}{5 \cdot A_1} \right)^2}{\left( \frac{V_2}{5 \cdot A_2} \right)^2}$$

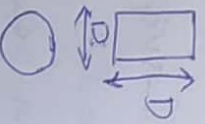
$$t_2 = t_1 \cdot \frac{V_2^2}{(5 \cdot A_2)^2} \times \frac{(5 \cdot A_1)^2}{V_1^2}$$

=

$$= \underline{\underline{40 \text{ sec}}}$$

? A steel slab having dimension  $30 \times 20 \times 5$  cm is produced by casting by using a side riser which is cylindrical in shape & diameter & height of riser is equal to D. what is the freezing ratio of mould?

$$\text{Freezing ratio} = \frac{(5A \cdot V_c)}{(SA \cdot V_r)}$$



Assignment?

A cube shaped casting 'x' solidifies in 5 min. what will be the solidification time for another cube shaped casting of the same metal & which is 8 times heavier than the original casting

$$= \frac{2((30 \times 20) + (20 \times 5) + (30 \times 5))}{\frac{2 \times \pi D^2/4 + \pi D^2}{\pi/4 D^3}}$$

$$= \frac{8D}{75} //$$

$$= \frac{8D}{75} //$$

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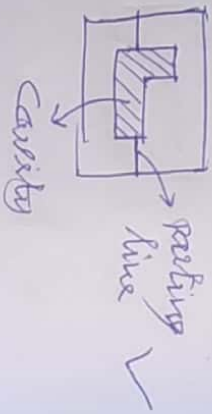
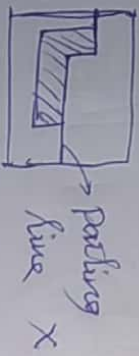


## Module-2:

### Product design consideration for

Defect free casting:

1. compensating shrinkage of molten metal by giving allowance
2. Parting line of the mold should be in single plane



3. Draft angle should be given
4. Riser should be located in heavier section
5. Abrupt change in sections should be replaced by fillets and tapers
6. The interior walls of component should be 20% thinner than exterior walls.



7. Hole placed in stressed areas should be given additional reinforcement
8. To minimize residual stresses in gear, pulley etc, a balance in the section designs are preferred

### Product design for economical casting:

1. Optimum product design
2. Optimal gating design
3. Simulation of mold filling, solidification & casting defects.
4. Optimizing process parameters for molding sand
5. Defect/minor control - Thermal stress control, optimizing heat treatment of molds to control distortion, warping)
6. Efficient cooling rate

### Shell casting/Molding:

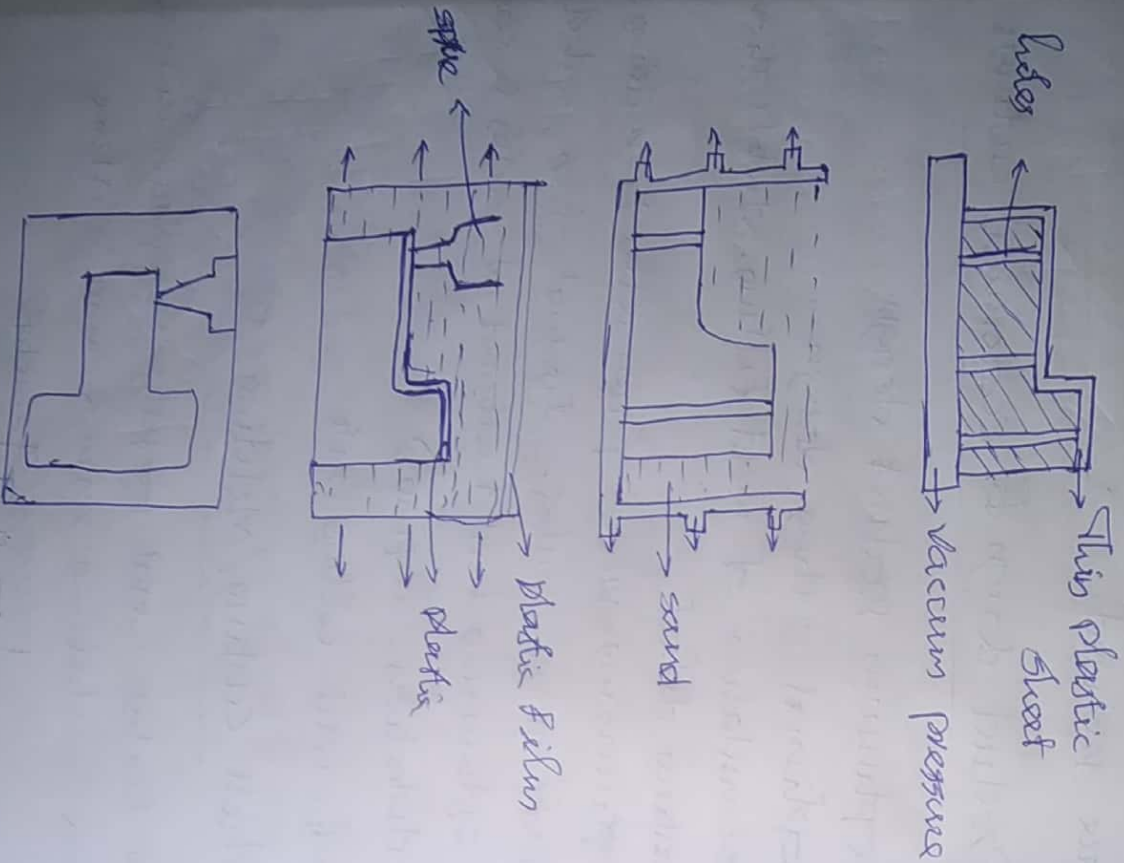
1. A heated metal pattern is placed over a box containing resin & sand mixture



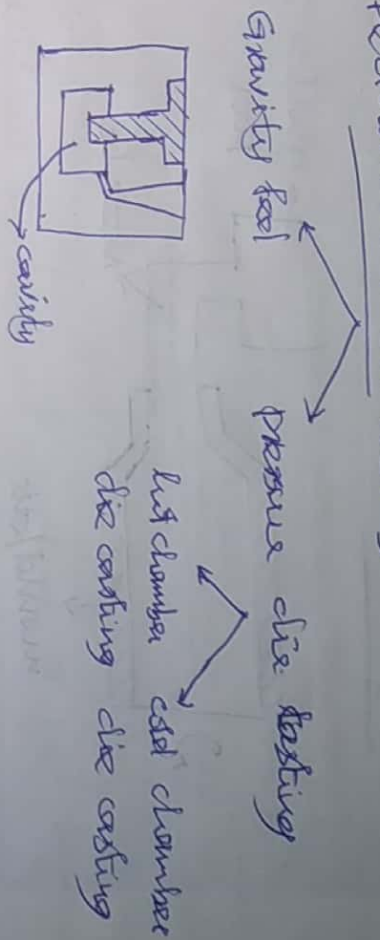


## 2 Investing the plate

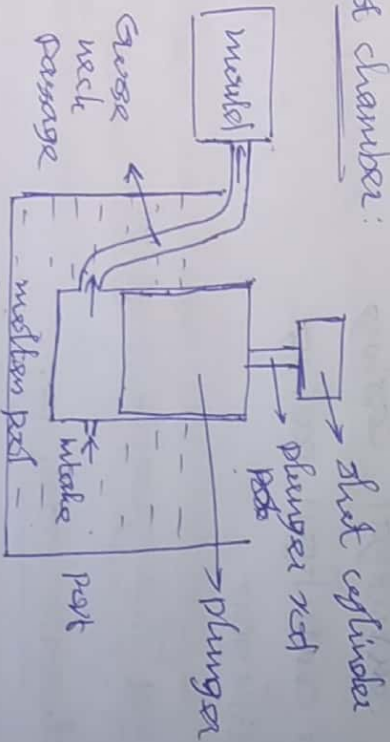
### Vacuum casting / Molding:



### Permanant mold casting:

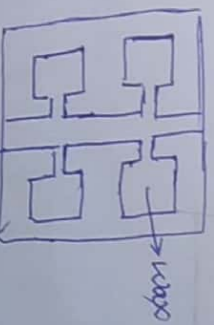


### Hot chamber:

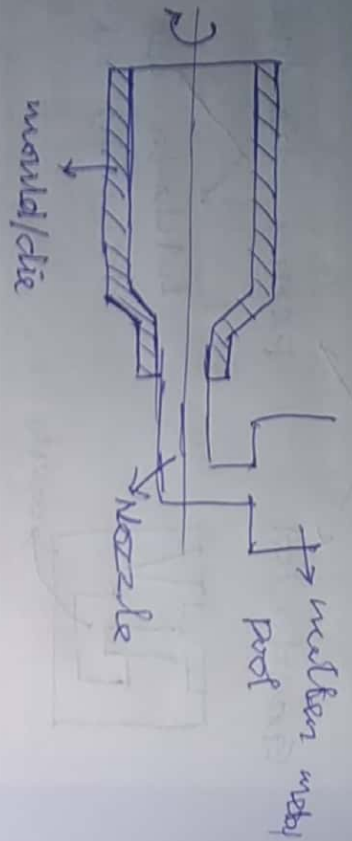


### Wax casting / Investment casting:

1. creating wax pattern
2. dipping wax pattern in a slurry of refractory material

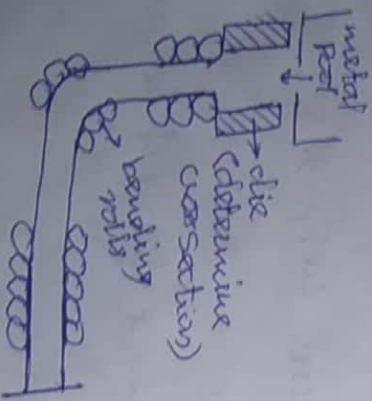


## Centrifugal casting:



1. True centrifugal casting
2. Semi centrifugal casting
3. Centrifuging
- 2 we get high quality product

## Continuous casting:



Billots, Rods

## Casting defects:

### 1. Pinhole:

These are tiny holes about 2mm diameter usually found in upper portion of mould. They usually appear in large number either at least surface of casting or just below the surface. It happens due to poor bending of mould.



### 2. Blowhole:

These are larger cavities than pinholes & it occurs inside the surface of casting which can be determined only by testing methods like ultrasonic, magnetic or x-ray analysis.

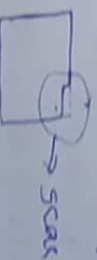


### 3. Open leak:

They are blowholes appearing on surface of casting & it's caused by poor bending, insufficient drying of mould & core.

### 4. Scal:

Shallow blows that appear on a flat surface





### 8 Roller:

There are sand covered with thin metal



### 9 Shrinkage cavity:

It is a depression which occurs during solidification process shrinkage cavity appear as angular edges compared to round surface of the porosity.

### 7 Cuts & whisker:

There are areas of excess metal. There defects appear when molten metal meets moulding sand.

A cut appear as low projection along the surface of clay phase which is in height as it extends from one side of the casting to other.

→ These defects can be prevented by designing the gating system properly, improving mould & core strength, adding more binder to facing & core sand.

### 4 Fusion:

This occurs when the sand grains fuse with molten metal. causes of fusion defects are lower bond strength, lower refractoriness of sand, too high pouring temp of molten metal.

### 9 Run out:

It is a defect when liquid metal falls out the mould leading to an incomplete casting. It can be prevented by designing the casting mould with provision. Also high temp can lead to wear & tear of mould.

### 10 Swell:

There are enlargement of casting particularly on vertical sprue caused by improper or soft ramming of mould.

### 11 Drop:

They occur when pieces of sand fall into the molten metal & appear as irregularity.

shaped projection on the top surface of the casting.

It can be prevented by using sand of higher strength, better ramming

## 12 Cold shut:

It is a type of surface defect where a line or crack is visible on casting surface. It happens when 2 streams of molten metal meet each other & lower temp will prevent the fusion.

Best way to prevent it is to increase the fluidity of molten metal

## 13 Mismatch:

It occurs when liquid metal is too cold to flow to the extremities of the mold before freezing.

Mismatches are characterized by incomplete filling of mold cavity

## 14 Shift:

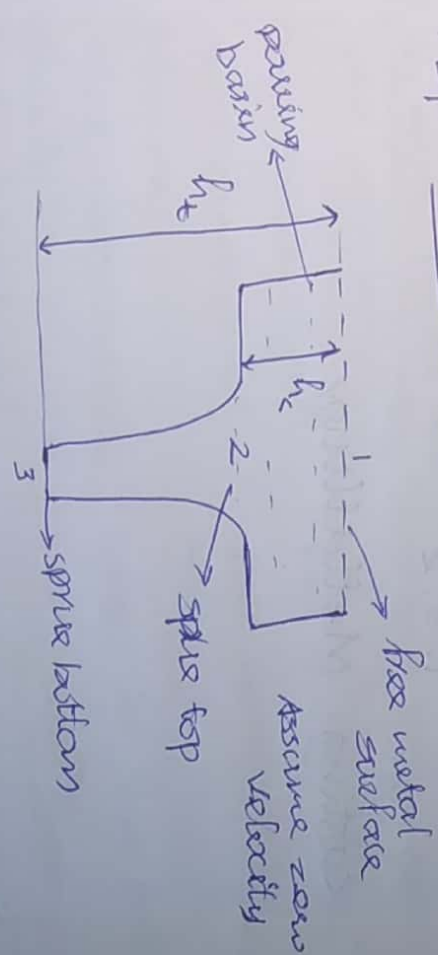
It is a defect caused due to misalignment of top & cheng part of mould. It is usually

Shifting are also visible in core. The causes of this defect could loose box pins, inaccurate pattern of chell pins

## 15 Flash:

This defect is seen as thin sheet of metal which are coming out of the parting planes. Flash may occur due to top crack or gap on core surface, insufficient weight of ~~flask~~ <sup>pouring</sup> & flask on the mould & improper

## Aspiration effect of sprue:



continuity eqn b/w ② & ③

$$A_2 V_2 = A_3 V_3$$

$$\frac{A_2}{A_3} = \frac{V_3}{V_2} = \frac{\sqrt{2gh_t}}{\sqrt{2gh_c}}$$

$$\frac{h_t}{h_c} = \left( \frac{A_2}{A_3} \right)^2$$



## Gating ratio:

a:b:c

a - sprue cross section area

b - runner cross section area

c - Gate cross section area

→ For pressurised casting

3:2:1

→ Non pressurised

1:3:2

## Casting Metallurgy



$$\left[ \left( \frac{h_1}{h_2} \right) - \frac{h_1}{h_2} \right]$$

$$\frac{1.49 \times 10^{-4}}{1.49 \times 10^{-4}} = \frac{1.49}{1.49} = 1$$